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Knowledge Management & Discovery Lab

Exercise 10 – Additional Practice

1. Silhouette coefficient.

Given are 7 two-dimensional points with coordinates shown in Table 1. The clustering result is:

$$G_1 = \{s_1, s_2\}, \quad G_2 = \{s_3, s_4, s_5\}, \quad G_3 = \{s_6, s_7\}.$$

Calculate the silhouette coefficient for the cluster G_2 . Use Euclidean distance as the distance measure. A partially completed distance matrix is given in Table 2. Complete any missing distances you need. Briefly interpret the result.

For a point $x \in X$, use

$$a(x) = \frac{\sum_{u \in X \setminus \{x\}} d(x, u)}{|X| - 1}, \quad b(x) = \min_{Y \neq X} \frac{\sum_{u \in Y} d(x, u)}{|Y|},$$

and

$$\text{silhouette}(x) = \frac{b(x) - a(x)}{\max(a(x), b(x))}.$$

Table 1: Point coordinates for the silhouette coefficient task.

ID	s_1	s_2	s_3	s_4	s_5	s_6	s_7
x	1	2	5	6	3	7	8
y	1	2	5	6	2	5	6

Table 2: Partially completed Euclidean distance matrix. The matrix is symmetric, and the diagonal is ignored.

d	s_1	s_2	s_3	s_4	s_5	s_6	s_7
s_1	–	□	5.66	7.07	2.24	□	□
s_2		–	4.24	□	1.00	□	□
s_3			–	1.41	□	2.00	□
s_4				–	□	□	2.00
s_5					–	5.00	□
s_6						–	□
s_7							–

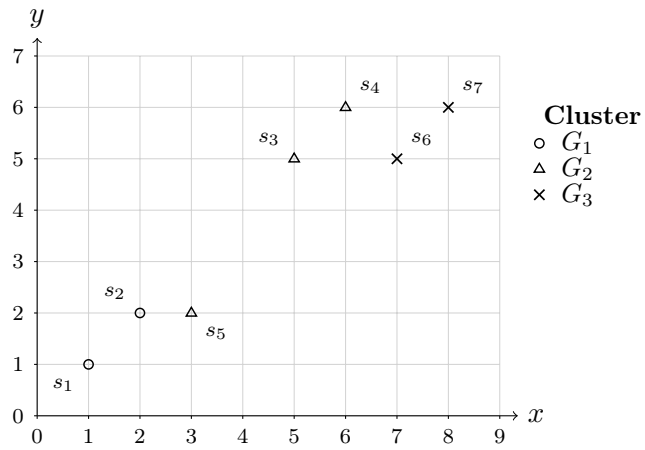


Figure 1: Clustering result for the silhouette coefficient task.